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Exploring the metabolic characteristics of fresh-cut sweet potatoes at different storage times based on untargeted metabolomics

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p-Coumaric acid (PubChem CID: 637542)
Peonidin (PubChem CID: 164544)
Naringenin (PubChem CID: 439246)
Kaempferol (PubChem CID: 5280863)
Procyanidin B5 (PubChem CID: 124017)

ABSTRACT

Sweet potato (*Ipomoea batatas* (L.) Lam.) is an important tuber crop. This study employed LC-MS/MS to investigate metabolic responses in fresh-cut purple sweet potatoes during short-term storage (0–5 d). Fresh cutting and storage jointly reprogrammed metabolic networks, with both basal metabolite levels and their dynamic changes closely linked to browning and quality deterioration. Amino acids, organic acids, and phenolic compounds were the major responsive classes. L-phenylalanine rapidly accumulated after cutting and remained stable, while L-isoleucine increased continuously, showing a positive association with wound response and browning mitigation. In contrast, several antioxidant phenolic acids and flavonoids significantly declined, suggesting that their depletion weakened antioxidant capacity and was negatively correlated with browning. KEGG analysis showed significant enrichment in phenylpropanoid biosynthesis, flavonoid biosynthesis, and phenylalanine metabolism. These metabolites may serve as potential markers for evaluating storage stability and browning sensitivity in fresh-cut purple sweet potatoes.

1. Introduction

Sweet potato [*Ipomoea batatas* (L.) Lam.] is one of the major food crops worldwide and is cultivated in all tropical and subtropical regions, particularly in Asia, Africa, and the Pacific (El Sheikha & Ray, 2017). Because purple sweet potatoes are rich in dietary fiber, polyphenols, anthocyanins, and a variety of functional amino acids, they are considered to have both nutritional value and health-promoting potential and have shown good application prospects in the field of ready-to-eat vegetables and functional foods (de Aguiar Cipriano et al., 2022). With the shift in consumption patterns toward "high nutrition-high convenience," fresh-cut sweet potatoes have gradually attracted attention as a product form suitable for ready-to-eat and light processing (Ölmez & Kretzschmar, 2009). However, cutting inevitably damages tissue structure, triggers a typical wound stress response, and leads to browning, texture deterioration, and flavor degradation, thereby significantly shortening the product's shelf life and reducing consumer acceptance (Liu, Hai, et al., 2025; Liu, Li, et al., 2025).

When sweet potato tissues are cut, the integrity of cellular structures and subcellular compartmentalization is disrupted, leading to severe perturbations of the original intracellular homeostasis and the rapid induction of wound responses characterized by oxidative stress. Previous studies have shown that mechanical injury can markedly enhance the generation of reactive oxygen species (ROS) and, through signal transduction pathways, drive rapid reprogramming of the metabolic network, shifting primary metabolism toward defense-related secondary metabolic pathways (Jacobo-Velázquez et al., 2015). During this process, the phenylpropanoid pathway is significantly activated, promoting the synthesis and accumulation of various phenolic compounds and their derivatives to buffer oxidative stress induced by mechanical damage (Guan et al., 2020; Marzec, 2025). Meanwhile, wound stress can further promote the reprogramming of primary to secondary metabolism via signaling pathways, resulting in a pronounced enhancement of phenylpropanoid metabolism and the biosynthesis of related phenolic

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compounds (Xu et al., 2023). Although some phenolic compounds exhibit low basal levels or are even undetectable in intact tissues, their biosynthesis is markedly induced after cutting. These newly synthesized metabolites not only participate in regulating tissue color changes and redox status but may also contribute to wound sealing and tissue repair through lignification or cell wall reinforcement, thereby playing important roles in quality evolution of fresh-cut tissues (Gao, Wang, et al., 2024; Gao, Yang, et al., 2024). Wound kinetics studies have further shown that certain metabolite families, such as lysophospholipids, jasmonate-related metabolites, and phenolic compounds, undergo significant changes, indicating that differences in oxidative sensitivity may be associated with variations in plant metabolomic profiles (Garcia et al., 2016; Li et al., 2024).

In recent years, high-throughput analytical methods based on LC-MS/MS have been widely applied to the study of post-harvest physiology and wound response mechanisms in fruits and vegetables (Widjaja et al., 2025). Related studies have demonstrated that metabolic changes in sweet potato during postharvest storage involve multiple metabolite classes, including amino acids, organic acids, phenolic acids, flavonoids, and sugar-derived metabolites, with many exhibiting pronounced time-dependent dynamics (Wan et al., 2025). In crops such as broccoli, lettuce, potato, and apple, phenylpropanoid-related phenolic compounds and their derivatives have been identified as potential metabolic biomarkers for browning sensitivity after cutting (Guan et al., 2020; Ouyang et al., 2025; Wills & Li, 2016). However, compared with studies focusing on metabolic changes in sweet potato under long-term storage conditions, systematic analyses of wound-responsive metabolic networks during short-term storage of fresh-cut sweet potato remain limited, particularly those linking basal metabolite levels with subsequent browning development or quality deterioration.

Based on this background, this study used purple sweet potato as the research material and employed untargeted metabolomics to systematically elucidate the metabolic response characteristics of fresh-cut sweet potato during short-term storage (0–5 d). By integrating multivariate statistical analysis with metabolic pathway enrichment analysis, key metabolites closely associated with fresh-cut processing and storage

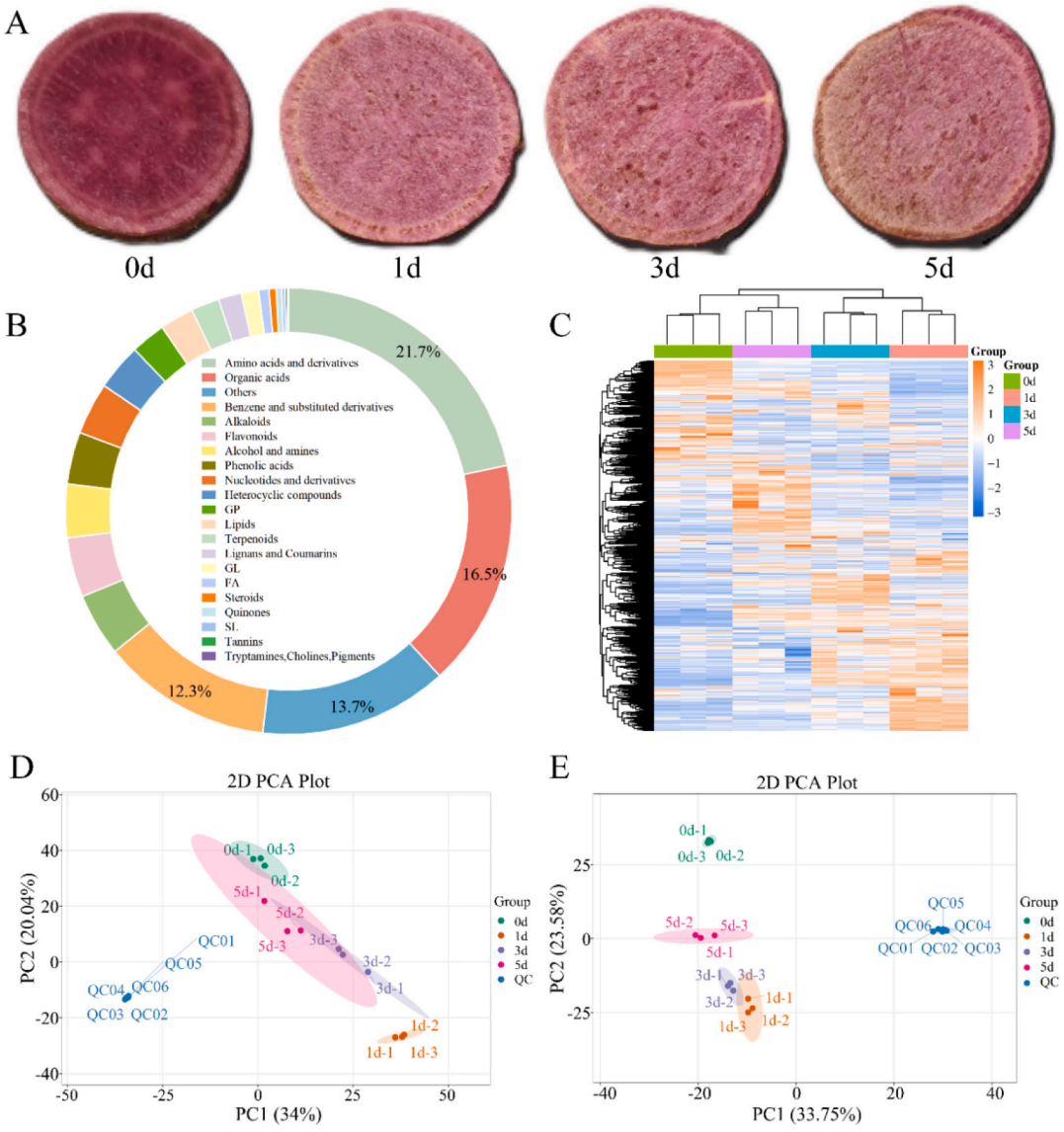


Fig. 1. Heatmap analysis and identification of metabolites in sweet potato flesh during different storage periods. A: Representative photographs showing the flesh color of sweet potato at different storage times. B: Composition analysis of the identified metabolites. C: HCA of total metabolites across the four storage stages. Each row represents an individual metabolite, and each column represents a sample. The color scale indicates the relative abundance of each metabolite, with red representing higher relative abundance and blue representing lower relative abundance. D: PCA score plot in positive ionization mode. E: PCA score plot in negative ionization mode. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

time were identified, with particular emphasis on the dynamic changes of amino acids, organic acids, and phenolic compounds. Furthermore, their potential relationships with browning development and quality deterioration in fresh-cut sweet potato were explored, providing metabolomics-based theoretical support for processing optimization, quality control, and raw material selection of fresh-cut sweet potato.

2. Materials and methods

2.1. Plant material

The purple sweet potato cultivar used in this study was "Zifei" (Fig. 1A), which was provided by the Enshi Tujia and Miao Autonomous Prefecture Academy of Agricultural Sciences. Samples were collected from the experimental base of the College of Forestry and Horticulture, Hubei Minzu University (30°17′ N, 109°29′ E, altitude 495.5 m). Tuber roots with uniform morphology and no visible mechanical damage were selected, sliced evenly to a thickness of 5 mm, and stored in a refrigerator. Four storage durations after fresh cutting were set: 0 d (fresh control), 1 d, 3 d, and 5 d. Storage conditions were maintained at 4 °C with a relative humidity of 60–70%. Samples were placed on open ceramic trays and stored under non-sealed conditions, without the use of modified atmosphere packaging or any regulation of gas composition (O_2/CO_2).

At the end of each storage period, the peel tissue was removed. The flesh tissue was then chopped and immediately frozen in liquid nitrogen to terminate enzymatic activity. After thorough homogenization, samples were aliquoted into 2 mL cryogenic tubes, rapidly frozen in liquid nitrogen, and transferred to a − 80 °C ultra-low temperature freezer for storage until untargeted metabolomics analysis. Three biological replicates were prepared for each treatment.

2.2. Dry sample extraction

The biological samples were first lyophilized using a vacuum freeze-dryer (Scientz-100F). The freeze-dried material was subsequently ground into a fine powder using a grinder (MM 400, Retsch) at 30 Hz for 1.5 min. Precisely 30 mg of the resulting powder was weighed on an analytical balance (MS105DM). To this, 1500 μL of a pre-cooled (−20 °C) 70% methanol aqueous solution, containing 2-chlorophenylalanine (purity ≥98%) as an internal standard at a working concentration of 1 mg L^{-1}, was added. (For sample amounts less than 30 mg, the extractant volume was added proportionally, maintaining a ratio of 1500 μL per 30 mg of sample). The mixture was subjected to intermittent vortexing, consisting of 30-s pulses repeated every 30 min for a total of six cycles. Following centrifugation at 12,000 rpm for 3 min, the supernatant was carefully collected. It was then filtered through a 0.22 μm microporous membrane into an injection vial for subsequent UPLC-MS/MS analysis.

2.3. LC-MS analysis conditions

Chromatographic separation was performed using a Waters ACQUITY UPLC HSS T3 column (1.8 μm, 2.1 × 100 mm) maintained at 40 °C. The mobile phase consisted of (A) water with 0.1% formic acid and (B) acetonitrile with 0.1% formic acid, delivered at a flow rate of 0.40 mL min^{-1} The injection volume was 4 μL. The gradient elution program was set as follows: initial conditions were 95% A and 5% B. The proportion of B was increased linearly to 65% over 5 min, then to 99% over 1 min, and held for 1.5 min. Subsequently, the mobile phase composition was returned to the initial conditions (95% A, 5% B) within 0.1 min, and the system was re-equilibrated for 2.4 min.

2.4. Mass spectrometry conditions

Mass spectrometric data were acquired in information-dependent acquisition (IDA) mode controlled by Analyst TF 1.7.1 software (Sciex, Concord, ON, Canada). The ion source parameters were as follows: ion source gas 1 (GS1), 50 psi; ion source gas 2 (GS2), 60 psi; curtain gas (CUR), 35 psi; source temperature, 550 °C; declustering potential (DP), ±80 V; and ion spray voltage floating (ISVF), ±5500 V (positive mode) / -4500 V (negative mode). For the TOF-MS survey scan, the mass range was set to 50–1250 Da with an accumulation time of 200 ms, and dynamic background subtraction was enabled. For subsequent product ion scans, the mass range was 50–1250 Da with an accumulation time of 40 ms. The collision energy was set at ±30 V with a spread of 15 eV. Additional settings included: unit resolution; charge state, 1; intensity threshold, 100 cps; exclusion of isotopes within 4 Da; mass tolerance, ±50 mDa; and a maximum of 12 candidate ions monitored per cycle.

2.5. Qualitative and quantitative analysis of metabolites

The raw mass spectrometry data were converted to mzML format using ProteoWizard, followed by peak picking, alignment, and retention time correction with XCMS. Peaks missing in more than 50% of samples within any group were removed. Remaining missing values were imputed using k-nearest neighbors for partially missing data (<50%), while high-missing features were imputed using a conservative minimal-value approach to avoid artificial signal inflation. Metabolite annotation was performed using a hierarchical identification strategy with explicit priority ranking, integrating an in-house MS/MS database (MS2-search-local, Metware Biotechnology, China) as the primary reference, followed by public spectral databases including METLIN, HMDB, KEGG, MoNA, and MassBank, and further complemented by in silico fragmentation prediction and network-based annotation (MS2-Insilico and metDNA workflow) to improve coverage of unidentified features. For features lacking MS/MS spectra, putative annotations were assigned based on accurate mass matching (MS1 level). Identification was performed using unified tolerances of 25 ppm for precursor ions, 50 ppm for MS/MS fragments, and 6 s for retention time alignment, and only metabolites with MS/MS similarity ≥0.3 and a final confidence score ≥ 0.5 were retained. The final identification score was calculated as a weighted combination of fragment matching (0.1), forward spectral similarity (0.3), and reverse spectral similarity (0.6), thereby prioritizing spectral consistency and database reliability. To correct systematic analytical variation, peak intensities were normalized using a support vector regression (SVR)-based QC correction model, constructed on pooled quality control (QC) samples injected periodically and applied in a feature-wise manner to model signal intensity drift over analytical time. The effectiveness of normalization was assessed using multiple QC-based criteria, including relative standard deviation (RSD) distribution, PCA-based clustering stability of QC samples, and inter-QC correlation analysis, and features with QC-based RSD >50% were excluded to ensure data robustness. Data from positive and negative ionization modes were combined, retaining, for each metabolite, the feature with the highest identification confidence and lowest variability to ensure annotation reliability.

2.6. Statistical analysis

This study used variable importance in projection (VIP) > 1, false discovery rates (FDR) <0.05, and fold change (FC) ≥ 2 or ≤ 0.5 as criteria for screening differential metabolites. Among these, VIP values were used to evaluate the contribution of each metabolite to the PLS-DA model, thereby reflecting its discriminative power between groups. FC was employed to quantify the magnitude of changes in metabolite levels across groups, with the defined thresholds indicating at least a two-fold upregulation or downregulation. To reduce false positives arising from multiple comparisons, *p*-values were adjusted using the Benjamini–Hochberg method, and metabolites with FDR < 0.05 were considered statistically significant. Subsequently, functional annotation and classification of the identified differential metabolites were performed using

the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. All data processing, statistical analyses, and visualizations were conducted using R software (version 4.5.1).

3. Results

3.1. Metabolic profile analysis

In this study, LC-MS/MS-based untargeted metabolomics was applied to investigate metabolic alterations in sweet potato following fresh cutting during different storage periods. To assess the reproducibility of metabolite extraction and instrumental detection, total ion chromatograms (TICs) of quality control (QC) samples were overlaid for comparison. The results indicated that retention times and peak intensities of identical samples were highly consistent across different analytical runs, demonstrating good signal stability (Figs. S1A, S1B). Pearson correlation analysis further showed that the correlation coefficients (|r|) among QC samples were close to 1, confirming the robustness and reliability of the analytical workflow **(Figs. S1C, S1D)**. In addition, data dispersion was evaluated using the coefficient of variation (CV), and more than 75% of metabolites in QC samples exhibited CV values below 0.3, further supporting the high stability and reproducibility of the metabolomics data **(Figs. S1E, S1F)**.

Following quality control filtering, 3415 metabolites were successfully identified, spanning a wide range of metabolite classes (Fig. 1B; **Table S1)**. The predominant categories included amino acids and derivatives (741 metabolites, 21.70%), organic acids (563, 16.49%), benzene and substituted derivatives (421, 12.33%), alkaloids (156, 4.57%), flavonoids (148, 4.33%), alcohols and amines (132, 3.87%), phenolic acids (129, 3.78%), nucleotides and derivatives (129, 3.78%), and heterocyclic compounds (115, 3.37%). Hierarchical clustering analysis (HCA) based on global metabolite profiles revealed clear separation between freshly cut samples and those stored for different durations, indicating pronounced differences in metabolic composition during storage (Fig. 1C**)**.

3.2. PCA analysis of sweet potato metabolites at different storage times

As shown in **Fig. 1D** and **E**, QC samples were tightly clustered in the PCA score plots, indicating good stability and robustness of the analytical method. PCA was performed using a dataset containing 3415 annotated metabolites. The PCA score plots revealed a clear separation between the 0 d group ($n = 3$) and the fresh-cut sweet potato samples stored for 1, 3, and 5 d under both ESI+ (Fig. 1D**)** and ESI− (Fig. 1E**)** ionization modes, suggesting that fresh-cut storage treatment resulted in marked differences in metabolomic fingerprints compared with the non-stored (0 d) samples. In the positive ion mode, the first two principal components accounted for 34.00% (PC1) and 20.04% (PC2) of the total variance, with a cumulative contribution of 54.04%. In negative-ion mode, PC1 and PC2 accounted for 33.75% and 23.58% of the variance, respectively, for a cumulative explanation rate of 57.27%. Moreover, the tight clustering of QC samples in the PCA score plots further demonstrated the high consistency of their metabolic profiles, thereby confirming the good stability and reproducibility of the analytical procedure used in this study.

3.3. Differential metabolite analysis based on partial least squares discriminant analysis (PLS-DA)

PLS-DA was used to compare metabolite profiles in sweet potato across different storage periods, to evaluate the contribution of metabolites to intergroup discrimination. The PLS-DA model graph shows that in the comparison of 1 d vs. 0 d, 3 d vs. 0 d, and 5 d vs. 0 d, the metabolite groups of sweet potato show significant separation, indicating significant differences in their metabolic profiles **(Fig. S2)**. To validate the reliability of the PLS-DA models, a 200-fold permutation test was conducted, confirming their statistical robustness. Specifically, significant discrimination was observed between 0 d and 1 d samples ($Q^2 = 0.970$, $R^2X = 0.681$, $R^2Y = 1.000$; **Fig. S3A**), 0 d and 3 d samples ($Q^2 = 0.952$, $R^2X = 0.651$, $R^2Y = 1.000$; **Fig. S3B**), and 0 d and 5 d samples ($Q^2 = 0.938$, $R^2X = 0.646$, $R^2Y = 1.000$; **Fig. S3C**). No evidence of overfitting was detected in any of the models.

3.4. Effects of storage duration on metabolism in fresh-cut sweet potatoes

Differential metabolites were screened based on the criteria of VIP > 1, FDR <0.05 and FC ≥ 2 or ≤ 0.5. As a result, a total of 1152 differential metabolites were identified across the four storage stages of sweet potato **(Table S2)**. These metabolites were mainly classified as amino acids and derivatives (264, 23.00%), organic acids (187, 15.37%), benzene and substituted derivatives (146, 12.66%), flavonoids (59, 4.77%), alkaloids (48, 4.24%), nucleotides and derivatives (47, 4.17%), and alcohol and amines (47, 3.98%), among others. Pairwise comparisons revealed distinct metabolic changes at different storage stages. Specifically, 735 differential metabolites were detected in the comparison of 1 d vs. 0 d, including 447 upregulated and 288 downregulated metabolites (Fig. 2A**)**. In the 3 d vs. 0 d comparison, 500 differential metabolites were identified, with 308 upregulated and 192 downregulated (Fig. 2B**)**. The 5 d vs. 0 d comparison yielded 363 differential metabolites, comprising 214 upregulated and 149 downregulated metabolites (Fig. 2C**)**. In addition, 190 differential metabolites (107 upregulated and 83 downregulated) were detected between 3 d and 1 d (Fig. 2D**)**. In comparison, 301 differential metabolites (144 upregulated and 157 downregulated) were identified in the 5 d vs. 1 d comparison (Fig. 2E**)**. Finally, 147 differential metabolites were observed between 5 d and 3 d, including 63 upregulated and 84 downregulated metabolites (Fig. 2F**)**. To characterize metabolite dynamics across storage stages further, an integrated analysis of all four time points was performed. A total of 151 metabolites were consistently and significantly altered when comparing the 0 d samples with stored samples (1 d, 3 d, and 5 d). These metabolites belonged to diverse chemical classes, including 25 amino acids and derivatives, 24 organic acids, 20 others, 18 benzene and substituted derivatives, 12 flavonoids, 9 phenolic acids, 7 nucleotides and derivatives, 7 heterocyclic compounds, 5 terpenoids, 5 alkaloids, 4 alcohol and amines, 3 lignans and coumarins, 3 GP, 2 tannins, 2 steroids, 2 lipids, 2 FA, and 1 GL (Fig. 2G**, Table S3)**.

Furthermore, multivariate statistical analysis was applied to evaluate metabolomic differences across the four storage stages comprehensively. Based on relative metabolite accumulation patterns, hierarchical clustering analysis (HCA) grouped the differential metabolites into four major clusters (Fig. 2H**)**. Metabolites in Cluster I exhibited higher accumulation levels at 0 d compared with 1 d, 3 d, and 5 d. Cluster II metabolites showed the highest abundance at 5 d, whereas Cluster III metabolites peaked at 3 d. In contrast, metabolites in Cluster IV reached their maximum accumulation at 1 d. These results indicate that metabolites exhibit distinct temporal accumulation patterns during post-cutting storage.

3.5. K-means clustering analysis

In order to systematically analyze the change characteristics of the relative content of metabolites between different treatment groups, the relative content data of all identified differential metabolites in the comparison group were scaled by unit variance, and K-means clustering analysis was further carried out. The K-means clustering results showed that the differential metabolites could be divided into six representative modules (Fig. 3A**)**, and the metabolites in each module exhibited similar changes across different treatment conditions. Subsequently, the top 20 metabolites with the highest differential multiples in each module were selected and analyzed. It was found that in subclass 1 (SC1), the accumulation level of compounds was lowest at 0 d. Combined with the relative abundance of differential metabolites, the levels of identified

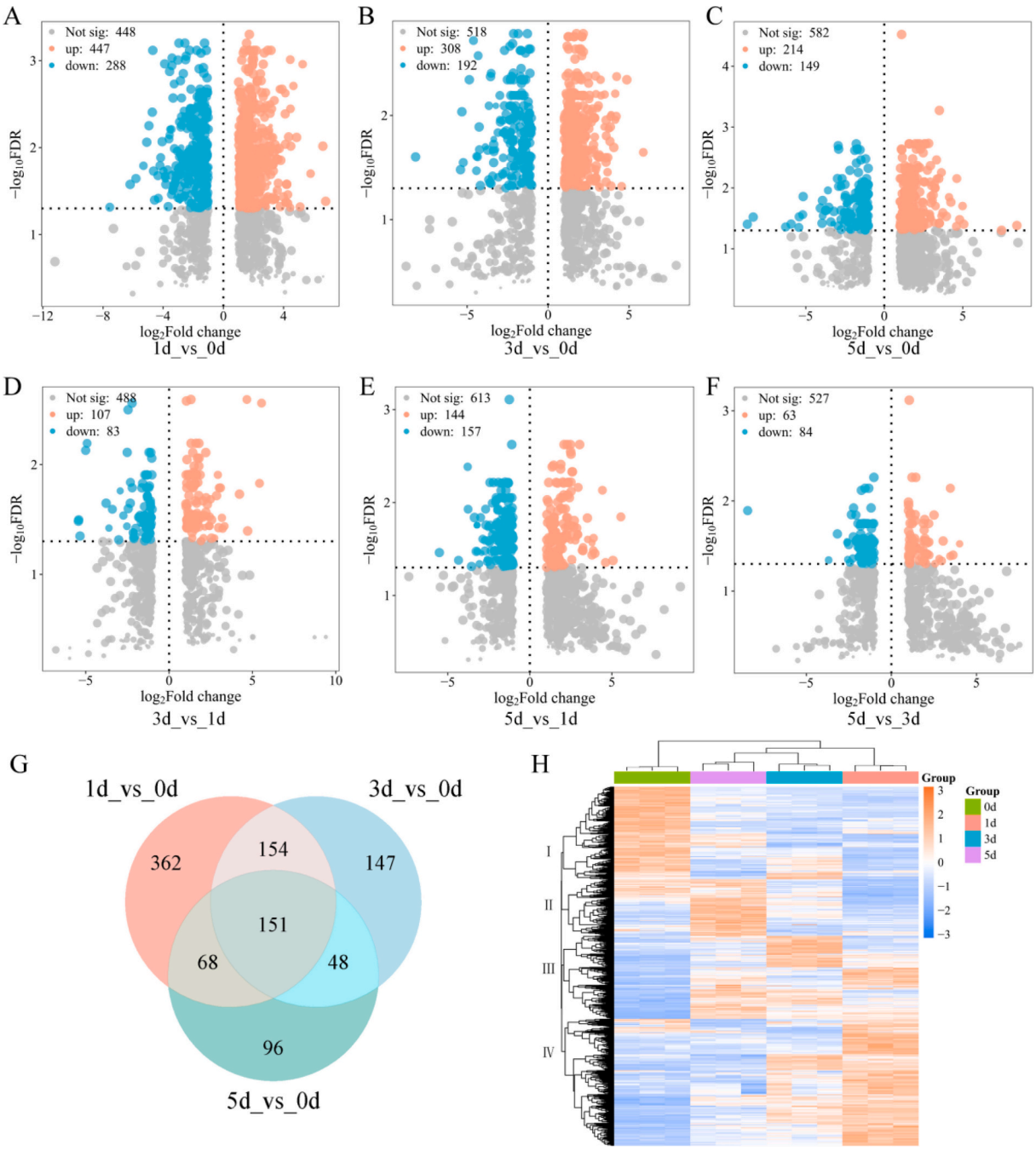


Fig. 2. Differences in metabolite levels in sweet potato flesh at different storage times. A-F: The volcano plots show the differential expression levels of metabolites in 1 d vs. 0 d, 3 d vs. 0 d, 5 d vs. 0 d, 3 d vs. 1 d, 5 d vs. 1 d, and 5 d vs. 3 d. Red dots indicate upregulated metabolites, while green dots indicate downregulated metabolites. G: The Venn diagram shows the overlap of differential metabolites in 1 d vs. 0 d, 3 d vs. 0 d, and 5 d vs. 0 d. H: HCA diagram of differential metabolites during different storage periods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

compounds such as cellobiotol, anhalamine, dethiosecoemestrin, and 10-hydroxycamptothecin in sweet potatoes were significantly increased after storage compared with those at 0 d. In subclass 2 (SC2) and subclass 5 (SC5), the contents of sulfazecin, 2,3′,4,6-tetrahydroxybenzophenone, vialinin A, procyanidin B5, carbamoylsulfanyl-acetic acid o-toluidide, prolylprolyllysine, and (−)-oleoside 11-methyl ester decreased significantly after storage. Notably, sulfazecin and procyanidin B5 exhibited 10–35-fold changes following storage. In subclass 3 (SC3), the accumulation levels of peonidin-3-O-β-galactopyranoside and forsythiaside at 1 d were higher than those at 0 d, 3 d, and 5 d. In subclass 4 (SC4), the accumulation level of compounds at 3 d was higher than that at 0 d, 1 d, and 5 d, among which the changes of 3,6-dimethylmangostin and 4-nitro-N-(quinolin-8-yl)benzenesulfonamide were particularly pronounced. In subclass 6 (SC6), the accumulation levels of compounds such as 2″-p-coumarylastragalin, carnitine, and albireodelphin A at 5 d were higher than those at 0 d, 1 d, and 3 d.

In addition, statistical analysis of the chemical categories of differential metabolites in each module found that organic acids, amino acids and derivatives, benzene and substituted derivatives, alkaloids, and flavonoids accounted for the main proportion in each clustering module (Fig. 3B), suggesting that these compound types have important metabolic response characteristics during sweet potato storage.

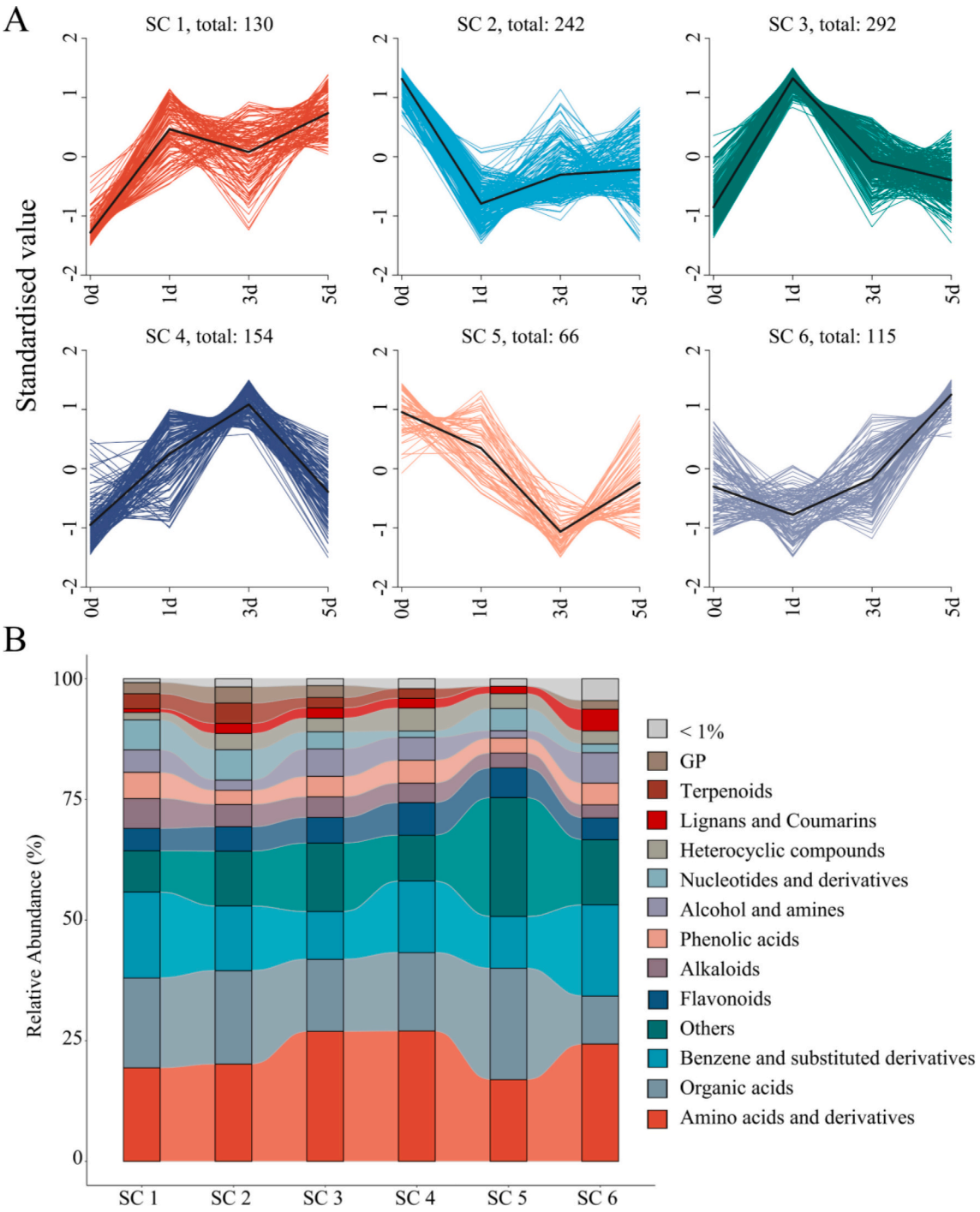


Fig. 3. Modular mapping based on differential metabolite trends and their chemical composition classification. A: Standardized abundance change trends of metabolites during storage periods at 0, 1, 3, and 5 days. Thin colored lines indicate the change trajectory of individual metabolites, while thick black lines represent the average change trend of metabolites within that cluster; the number of metabolites in the corresponding cluster is labeled in each subfigure title. B: Relative abundance distribution of metabolites by chemical classification within each module. Different colors represent distinct compound categories, including organic acids, amino acids and their derivatives, benzene and its substituted derivatives, alkaloids, phenolic acids, flavonoids, terpenoids, lignans, and coumarins, reflecting differences in metabolite composition under varying dynamic patterns.

3.6. Differential metabolite pathway analysis

To elucidate changes in sweet potato metabolic pathways across different storage periods, this study mapped metabolites from each comparison group to the KEGG database and evaluated the distribution characteristics of metabolites in each comparison group using KEGG functional enrichment analysis (Fig. 4A-E). In the early storage stage (1 d vs. 0 d), 79 pathways were significantly enriched, with prominent involvement of biosynthesis of nucleotide sugars, amino sugar and nucleotide sugar metabolism, and tryptophan metabolism, along with secondary metabolic pathways such as flavonoid and isoflavonoid biosynthesis (Fig. 4A). At 3 d compared with 0 d, 80 pathways showed significant enrichment, mainly associated with biosynthesis of secondary metabolites, nucleotide metabolism, tyrosine metabolism, and alkaloid biosynthesis (Fig. 4B). By 5 d of storage, the number of enriched pathways increased to 84, predominantly involving biosynthesis of

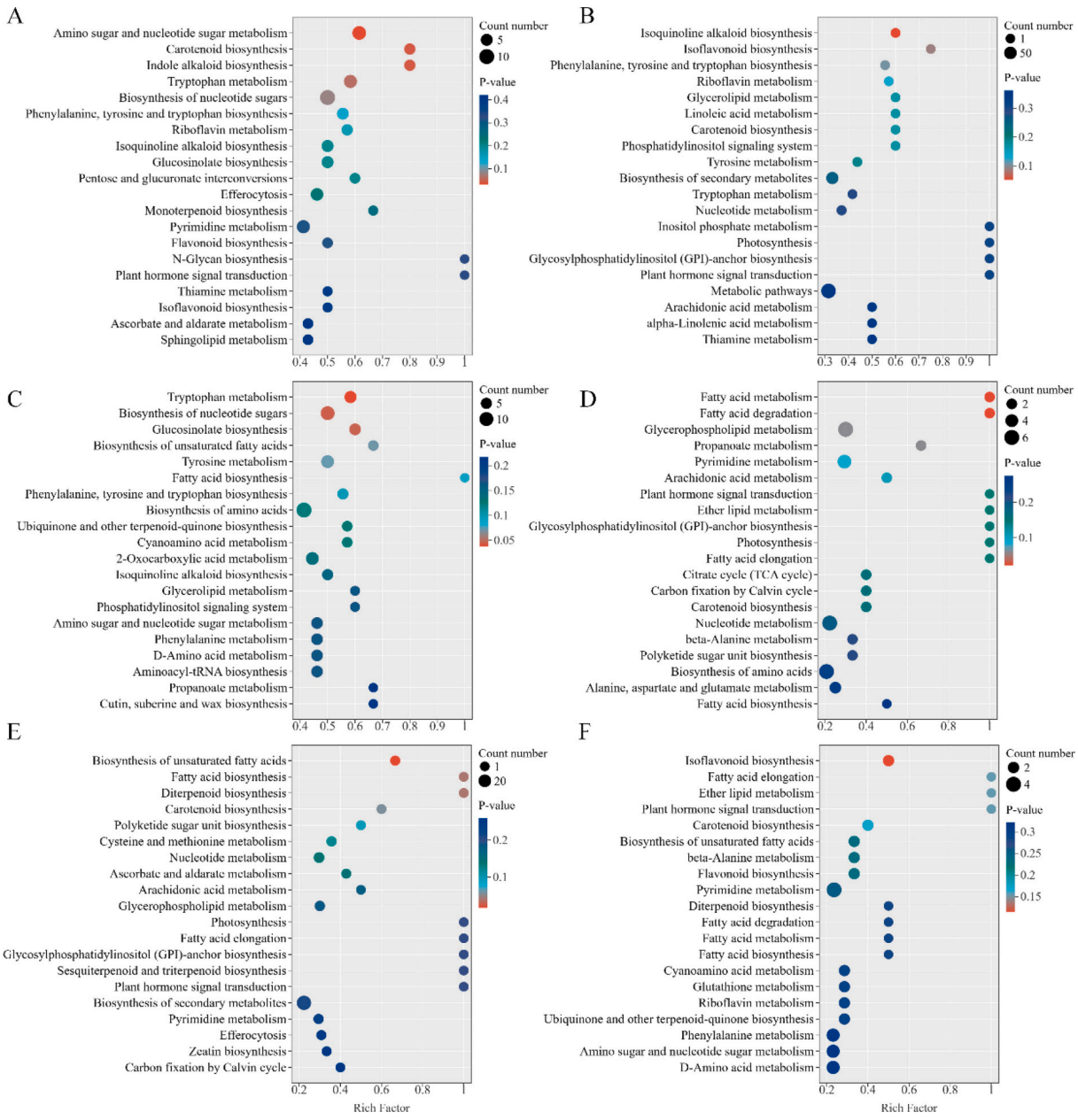


Fig. 4. KEGG enrichment plot of differential metabolites with A 1d vs. 0d, B 3d vs. 0d, C 5d vs. 0d, D 3d vs. 1d, E 5d vs. 1d, F 5d vs. 3d. The horizontal axis and vertical axis denote the enrichment factor and the corresponding metabolic pathway name for each pathway, respectively. The color of the dot indicates the *P*-value obtained from the hypergeometric test; The color ranges from blue to red, with a deeper red indicating lower *P*-values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

amino acids, nucleotide sugar biosynthesis, tyrosine metabolism, and 2-oxocarboxylic acid metabolism (Fig. 4C). Pairwise comparisons among stored samples further revealed stage-specific metabolic shifts. In the 3 d vs. 1 d comparison, 68 enriched pathways were mainly related to glycerophospholipid metabolism, nucleotide metabolism, amino acid biosynthesis, and pyrimidine metabolism (Fig. 4D). Similarly, 68 pathways were significantly enriched in the 5 d vs. 1 d group, with a strong enrichment in secondary metabolite biosynthesis, nucleotide metabolism, glycerophospholipid metabolism, and sulfur-containing amino acid metabolism (Fig. 4E). In the late storage comparison (5 d vs. 3 d), 60 pathways were significantly enriched, primarily including pyrimidine metabolism, amino sugar and nucleotide sugar metabolism, phenylalanine metabolism, and flavonoid-related pathways (Fig. 4F).

4. Discussion

Purple sweet potato is rich in dietary fiber, polyphenols, amino acids, and other bioactive nutrients, showing broad potential in ready-to-eat vegetables and functional food applications (Alam, 2021). With increasing consumer demand for highly nutritious and convenient foods, fresh-cut sweet potato has attracted growing attention due to its processing adaptability and ease of consumption. However, cutting inevitably causes tissue damage, triggering wounding stress and metabolic imbalance, thereby accelerating quality deterioration and shortening shelf life (Chen et al., 2023). Therefore, metabolite-level characterization of the dynamic responses of fresh-cut sweet potato is essential for understanding its quality formation mechanisms. In this study, LC-MS/MS-based untargeted metabolomics was applied to systematically investigate metabolic response profiles of fresh-cut sweet potato during

short-term storage (0–5 days). Multivariate statistical analyses (PCA, PLS-DA, and HCA) revealed a clear metabolic separation between fresh-cut samples (0 d) and storage stages (1–5 d), indicating that cutting is a key trigger driving metabolic reprogramming, while storage time further amplifies this shift. The number of differentially accumulated metabolites showed dynamic variation across comparisons, suggesting that metabolic responses transition from an initial rapid fluctuation phase to a relatively stable state over time. These results indicate that metabolic regulation in fresh-cut sweet potato is not a linear process but instead exhibits pronounced stage-dependent characteristics. It should be noted that browning-related physiological and biochemical parameters were not directly measured in this study. Therefore, the relationship between metabolite changes and browning remains correlative.

Amino acids, as critical nutritional and functional components in sweet potatoes, play critical roles in quality formation and health-promoting properties. In this study, four key essential amino acids were identified, including *L*-phenylalanine (Phe), L-isoleucine (Ile), L-histidine (His), and L-tryptophan (Trp). Among them, Phe showed a significant increase in all comparisons between fresh-cut treatment and post-cut storage, whereas no significant differences were observed among different storage durations. This pattern indicates that Phe content rapidly increases shortly after cutting and subsequently stabilizes, which is consistent with previous reports on Phe dynamics during potato storage (Davids et al., 2004). Phenylalanine is a key precursor in the phenylalanine metabolic pathway, and its accumulation may influence both wound responses and changes in nutritional value (He et al., 2024). The content of Ile gradually increased with extended storage duration, whereas His showed a significant decrease at 5 d compared with 0 d. This pattern of change is consistent with the dynamic changes in essential amino acids during the long-term storage of sweet potatoes (Ren et al., 2020). Previous studies have demonstrated that exogenous application of Ile can effectively alleviate browning in fresh-cut potato by suppressing polyphenol oxidase activity and enhancing antioxidant capacity (Meng et al., 2022). Therefore, the endogenous accumulation of Ile observed in this study may represent an essential regulatory mechanism by which sweet potato mitigates fresh-cut-induced browning. Meanwhile, the pronounced decrease in His during late storage may contribute to a reduction in His-associated bitterness, which is in line with previous observations on flavor changes (Wan et al., 2025). KEGG pathway enrichment analysis showed that tryptophan metabolism and phenylalanine, tyrosine, and tryptophan biosynthesis were highly enriched in comparisons involving fresh-cut treatment and storage, closely corresponding to the observed changes in these essential amino acids. In addition, previous studies have reported that mechanical injury in lettuce enhances phenylalanine ammonia-lyase (PAL) activity, thereby powerfully activating phenylpropanoid biosynthesis and promoting the synthesis and accumulation of soluble phenolic compounds (García et al., 2017). Similarly, significant enrichment of Phe in the phenylpropanoid biosynthesis pathway was detected in this study, suggesting that sweet potato and lettuce may share partially conserved metabolic regulatory mechanisms in response to wound-induced stress.

Mechanical fresh-cutting, as a typical form of wound stress, can rapidly redirect primary carbon metabolism toward secondary metabolism and induce the biosynthesis of phenolic compounds and related secondary metabolites with antioxidant functions, thereby buffering oxidative stress triggered by mechanical damage (Hu et al., 2022). Phenolic acids and flavonoids represent the predominant phenolic constituents in sweet potato, and their dynamic changes are closely associated with antioxidant capacity (Im et al., 2021). In this study, a total of 104 phenolic compounds showing significant differences were identified, including 45 phenolic acids and 59 flavonoids. The flavonoids comprised multiple subclasses, such as flavones, flavonols, isoflavones, anthocyanidins, and flavanones, reflecting the high complexity of phenolic metabolic responses in fresh-cut sweet potato. Further analysis revealed that the relative contents of 3-p-coumaroylquinic acid, leucodelphidin, and kaempferol 3-O-feruloyl-caffeoyl-sophoroside 7-O-glucoside, which were relatively high at 0 days, all decreased significantly after storage. Previous studies have reported that coumarate ester compounds possess potential antimicrobial and preservative properties (Wu et al., 2016). Leucodelphidin has been identified as a key contributor to the antioxidant properties of fruits (Li et al., 2025), while kaempferol derivatives with extensive glycosylation and acylation modifications generally exhibit strong antioxidant capacity (Choung et al., 2017; Deng et al., 2019; Zhang et al., 2025). Consistent with the K-means clustering results, procyanidin B5 in the SC2 module showed a marked decrease with storage and a large magnitude of change, suggesting that some key antioxidant substrates were continuously consumed during storage. This depletion may weaken the overall antioxidant capacity of sweet potato and promote the occurrence of fresh-cut browning. Forsythiaside, identified as a representative compound in the SC3 module, also exhibits antioxidant activity (Wang et al., 2025). Meanwhile, several flavonoids accumulated significantly during storage, suggesting that fresh-cut sweet potato may still counteract oxidative stress by inducing the synthesis of specific antioxidant metabolites (Hu et al., 2022). Among these, 3′,4′-dimethoxyflavone increased by 55.55-fold at 1 d compared with 0 d; this compound has been reported to possess antioxidant, anti-inflammatory, and neuroprotective activities (Fatokun et al., 2013; Kurowska & Manthey, 2004). KEGG enrichment analysis showed that flavonoids were mainly enriched in isoflavonoid biosynthesis and flavonoid biosynthesis. In contrast, phenolic acids were primarily enriched in phenylpropanoid biosynthesis, phenylalanine metabolism, and pathways related to secondary metabolite biosynthesis. These findings are highly consistent with previous studies on wound responses in fresh-cut fruits and vegetables (Hu et al., 2022).

Sweet potato is rich in various organic acids, including citric acid, pyruvic acid, quinaldic acid, gambogic acid, tropate, succinic acid, fumaric acid, and shikimic acid (Minemba et al., 2019; Wan et al., 2024). Although natural foods generally contain multiple organic acids, their flavor profiles are often dominated by one or two high-abundance acids, while the remaining acids are present at trace levels (Wu et al., 2020). Previous studies have shown that changes in the contents of citric acid or malic acid can significantly influence the flavor and sensory properties of natural foods (Gao, Wang, et al., 2024; Gao, Yang, et al., 2024; Liu, Hai, et al., 2025; Liu, Li, et al., 2025). In this study, citric acid maintained a relatively high abundance across all treatments but showed a significant decrease only at 3 d, with no significant difference detected between 0 d and 5 d. In contrast, D-malic acid also remained at a high level throughout storage but exhibited a significant increase during post-cut storage. Based on the dynamic variation patterns of the two compounds during storage, the fluctuation in citric acid was relatively limited, whereas D-malic acid showed a marked accumulation trend, suggesting that it may be an important responsive component in the changes of organic acid metabolism during the storage of fresh-cut sweet potato. Considering the potential role of malic acid in flavor regulation of foods, these results indicate that it may be involved in the alteration of flavor characteristics during storage; however, further confirmation through sensory evaluation is still required.

In addition to the comprehensive analysis of major metabolites, we identified six metabolic modules (SC1–SC6) based on K-means clustering, revealing that different classes of metabolites exhibit distinct time-dependent accumulation or depletion patterns during storage. Specifically, SC1 and SC3 primarily represented early metabolic responses to fresh-cut treatment, with representative compounds showing significant increases at 1 d after cutting. In addition to the phenolic compound forsythiaside, these modules included cellobiotol, and anhalamine. These results indicate that fresh-cut treatment not only markedly perturbs phenolic, amino acid, and organic acid metabolic pathways but also rapidly induces the accumulation of sugar-derived metabolites and specific secondary metabolites with potential antioxidant activity, thereby buffering oxidative stress induced by mechanical injury. This metabolic pattern is consistent with early defensive

metabolic activation observed during fresh-cut storage of melon (Guo et al., 2023). Alkaloids and terpenoids are considered important defensive metabolites that play a role in plant responses to various biotic and abiotic stresses (Kumar et al., 2025; Sha et al., 2024). In this study, naringenin and traumatic acid in the SC4 module reached peak levels at 3 d of storage, suggesting that during the mid-storage stage, metabolic regulation in fresh-cut sweet potato may gradually shift from initial wound responses toward sustained modulation of defense-related secondary metabolic pathways (Gouda et al., 2022; Kim et al., 2023). In addition, the significant accumulation of specific steroidal saponins, lipids, and aromatic compounds in the SC6 module at 5 d indicates that, with prolonged storage, the sweet potato metabolic network may undergo structural reorganization and progressively transition toward metabolic processes associated with senescence or tissue degradation. Meanwhile, several flavonoids in the SC5 module showed a significant increase 1 d after cutting, followed by a decline, further highlighting the marked dynamic plasticity and time-dependent regulation of metabolic processes in fresh-cut sweet potatoes.

5. Conclusion

Fresh-cut sweet potato exhibited highly dynamic and time-dependent metabolic regulation during short-term storage. Essential amino acids, such as Phe and Ile, together with phenolic metabolic pathways, played key roles in wound-stress responses, contributing to antioxidant defense while simultaneously influencing nutritional and flavor quality. Some highly active antioxidant substrates were continuously consumed during storage, whereas the induced accumulation of other flavonoids and defense-related secondary metabolites reflected a compensatory metabolic mechanism. Importantly, these metabolites may serve as potential biomarkers for monitoring the quality of fresh-cut produce and assessing its storage condition. For instance, changes in phenolics and amino acids may be used to assess oxidative status and nutritional quality, while D-malic acid could serve as an indicator of flavor changes during storage. Such metabolite-based indicators provide a basis for developing rapid detection tools and precision postharvest management strategies in the fresh-cut industry. However, it should be noted that this study is based on metabolomic analysis and does not directly assess browning phenotypes. Future research should integrate targeted metabolite validation, enzymatic assays, and direct quality measurements to establish causal links between metabolic changes and quality traits. Overall, this study reveals stage-dependent metabolic reprogramming and provides a foundation for future multi-omics integration to further elucidate postharvest regulatory mechanisms and support practical applications.

CRediT authorship contribution statement

Xiaolin Wan: Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Xiuzhi Wang:** Supervision, Investigation. **Moxian Chen:** Validation, Investigation. **Lingjun Cui:** Supervision, Investigation. **Qiang Xiao:** Writing – original draft, Supervision, Funding acquisition.

Informed consent statement

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fochx.2026.104066.

Data availability

Data will be made available on request.

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